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Hybrid-reluctance actuated fast steering mirror

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Summer Semester, 2026

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Contents

1 Introduction

Fast Steering Mirrors (FSMs) are specialized optical devices designed to precisely control the direction of light beams. These mirrors are pivotal in various high-speed optical systems, including laser communication, target tracking, and imaging stabilization platforms. FSMs operate by adjusting their angles with high precision and speed, allowing for rapid changes in the beam's trajectory. This capability is essential for applications requiring fine, dynamic adjustments to compensate for motion or to accurately position a laser beam.

The core of FSM's functionality lies in its actuators, which can be based on piezoelectric, electromagnetic, or other mechanisms, providing the dynamic torques and required range. These actuators enable mirrors to oscillate at a certain frequency for scanning applications or maintain stable positions for pointing applications, ensuring high-precision alignment even at high frequencies. FSMs are critical in the advancement of optical technologies, offering solutions for beam steering in optical communication systems, correcting aberrations in real-time in adaptive optics, and extending the measurement range of optical sensors. Their ability to high-dynamically and accurately steer beams of light makes them invaluable in both scientific research and commercial applications.

2 Fast Steering Mirror

The FSM for this course is driven by hybrid reluctance actuators, since they revealed the highest range-bandwidth product in FSMs and deliver higher forces per volume in comparison to Lorentz actuators, enabling a compact system design with high performance [1].

2.1 Hybrid reluctance actuator principle

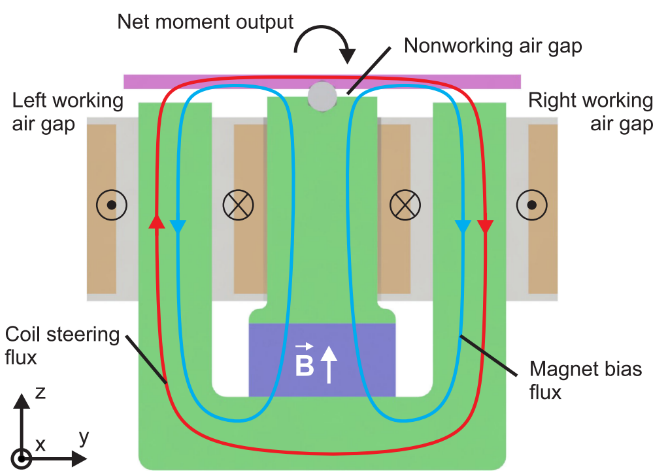


Figure 1: Working principle of a hybrid reluctance actuated FSM.

The working principle of hybrid reluctance actuators relies on the biasing flux of a permanent magnet (blue), as shown in Fig. 1. The bias flux travels over a nonworking air gap to the ferromagnetic mover and re-closes over the left and right working air gaps with the ferromagnetic stator (green), resulting in a static biasing magnetic flux. Due to the symmetric design, the non-working air gap does not contribute to the torque acting on the mover.

A current in the serially connected actuation coils yields a coil steering flux (red), in the left and right working air gaps. By super-positioning both magnetic fields, the flux in the right working air gap is increased and the flux in the left working air gap is decreased. The reluctance forces of each air gap are defined by

$$F_{left} = \frac{\Phi_{left}^2}{2\mu_0 A} \tag{1}$$

$$F_{right} = \frac{\Phi_{right}^2}{2\mu_0 A}, \tag{2}$$

with the magnetic fluxes in the left and right working air gap Φ_{left}, Φ_{left}, the permeability of vacuum μ_0 and the cross-section of the air gaps A. The magnetic fluxes in the working air gaps, can be expressed as:

$$\Phi_{left} = \Phi_{pm} - \Phi_{coil} \tag{3}$$

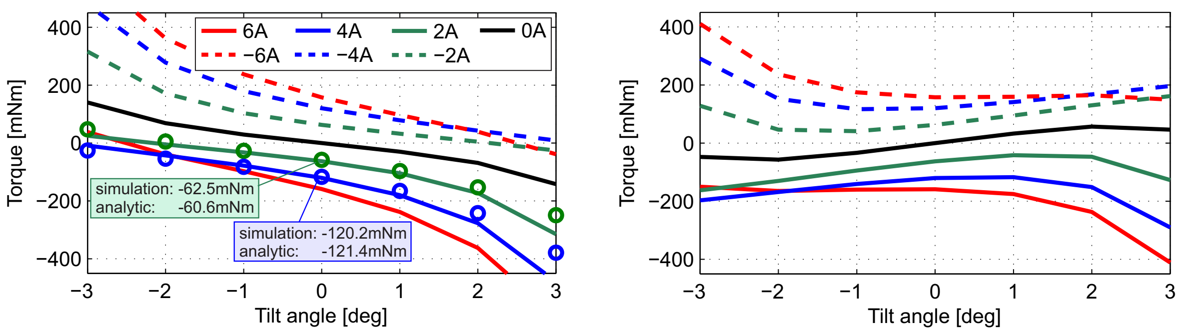
$$\Phi_{right} = \Phi_{pm} + \Phi_{coil}, \tag{4}$$

with the biasing flux of the permanent magnet Φ_{pm} and the coil steering flux Φ_{coil}. The actuation architecture can be described as pull/pull-configuration due to the pulling forces at each side of the mover. With the coil currents as input of the actuator, the reluctance force at each side of the mover can be differentially changed, resulting in the mover torque

$$\tau_{HRA} = \frac{F_{right} - F_{left}}{2} d_m \ , \tag{5}$$

with the mover diameter d_m as lever arm. The torque of the hybrid reluctance actuator can be linearly approximated in a sufficiently small region around an operating point with

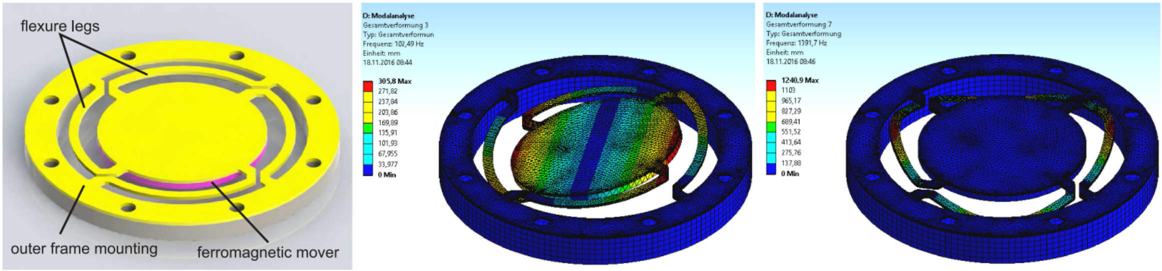
$$\tau_{HRA} \approx k_i i_{coil} + k_\Theta \theta \ , \tag{6}$$



(a) Torque of the hybrid reluctance actuator for various actuation currents without the flexure.

(b) Effective torque of hybrid reluctance actuator with the stabilizing flexure for various currents.

Figure 2: Torque of one axis of the tip/tilt FSM.



(a) (b) (c)

Figure 3: (a) Tip/tilt flexure of the FSM with the ferromagnetic mover mounted underneath and (b,c) the modal analysis of the flexure.

with the motor-constant $k_i = \frac{d\tau_{HRA}}{di_{coil}}$ and the rotational stiffness $k_\Theta = \frac{d\tau_{HRA}}{d\Theta}$. In this laboratory exercise, the linear approximation is sufficient to model the behavior of the actuator system at specific operating points.

Analytical results and finite element method simulations of the toque generated by one axis of the hybrid reluctance actuator used in the FSM are shown in Fig. 2a. Since the permanent magnet introduces a negative stiffness to the system, the FSM becomes open-loop unstable. To get an open-loop stable system, a flexure with a positive stiffness, which overcompensates the negative stiffness of the hybrid reluctance actuator, is added to the FSM design. The flexure of the FSM is depicted in Fig. 3. The effective torque of the hybrid reluctance actuator with the stiffness of the flexure is shown in Fig. 2b. A positive slope (positive effective stiffness) indicates the stable open-loop region of the FSM, which is around the center position of $\Theta = 0°$. This property enables the open-loop system identification of the FSM in this course. The system becomes open-loop unstable towards higher mover deflection angles.

2.2 FSM prototype

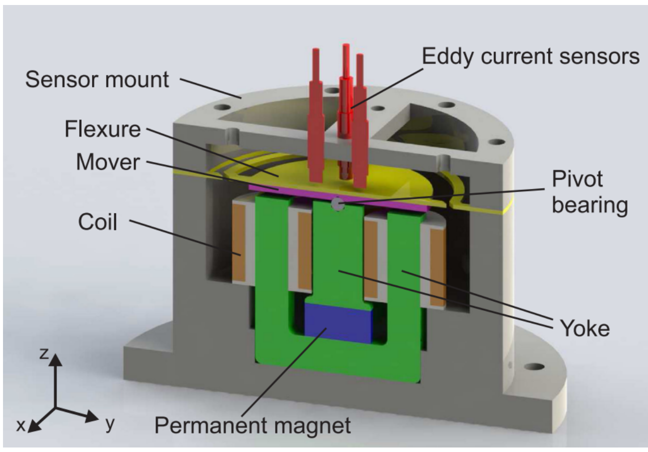


Figure 4: Cross section of the tip/tilt FSM with hybrid reluctance actuator mounted with eddy current sensor to identify the system behaviour [2].

To steer a laser beam in two degrees-of-freedom with the FSM, a tip/tilt flexure is designed to rotationally deflect the mirror around the pivot bearing in the orthogonal x- and y-axis, as shown in Fig. 4. Both actuator axes are designed equivalently on the basis of the hybrid reluctance actuation principle with one central permanent magnet for the biasing magnetic flux, to achieve an equal performance of the tip and tilt axis. Each axis of the FSM consists of two coils, which are connected in series.

Eddy current effects on the one hand decrease the energy efficiency of the FSM and on the other hand result in a phase loss of the system dynamics towards high frequencies. To minimize eddy current effects of the FSM, the ferromagnetic core of the stator is layered with a thickness of 0.5 mm, as visible in Fig. 5.

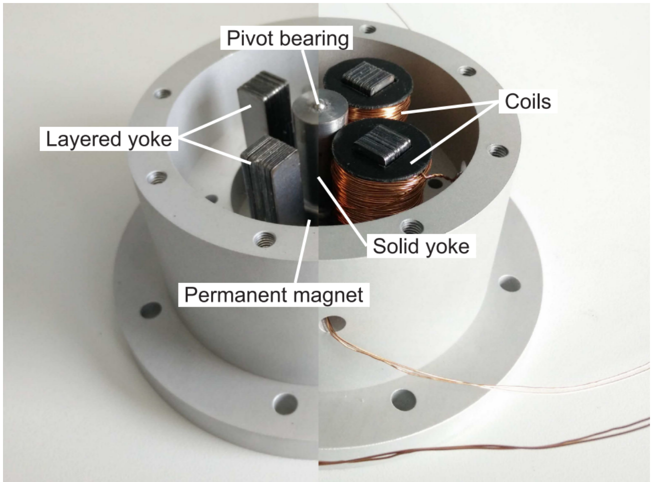


Figure 5: Layered ferromagnetic stator of the FSM with the acutation coils mounted at the right side.

The modal analysis of the flexure, depicted in Fig. 3b, shows the first mode shape of the tip axis at a frequency of about 100 Hz. Since the tip and tilt axis are symmetric, the second mode shape is equivalent to the first one, but rotated by 90 deg around the vertical axis. These two mode shapes are desired, as the mirror should be restricted to the tip/tilt movement. The third mode shape at about 1.4 kHz, shown in Fig. 3c, reveals the decoupling of the leaf springs. This decoupling of the leaf springs is undesired, since the flexure loses its capability of guiding the mover in the two desired rotational degrees of freedom.

3 Control system

One goal of this course will be to follow a disturbed object with a laser beam, therefore the position control of the FSM is required. The inputs of the FSM are the actuation currents I_x, I_y and the outputs are the angular mirror deflections of Θ_x, Θ_y of each axis, being a MIMO system (Multiple Input Multiple Output). Despite the orthogonal alignment of the axes, a cross-talk $G_{xy}(s)$ and G_{yx} between both axes is present e.g. due to the coupled magnetic circuits or a misalignment of the position sensor. However, the cross-talk should be kept rather small by virtue of the mechatronic system design and therefore two SISO (Single Input Single Output) controllers $C_P(s)$ are sufficient for this laboratory exercise.

The electrical subsystems $A(s)$, consisting of the coil inductance and resistance, are controlled by analog current control circuits with a bandwidth of about 4 kHz. Due to the inner

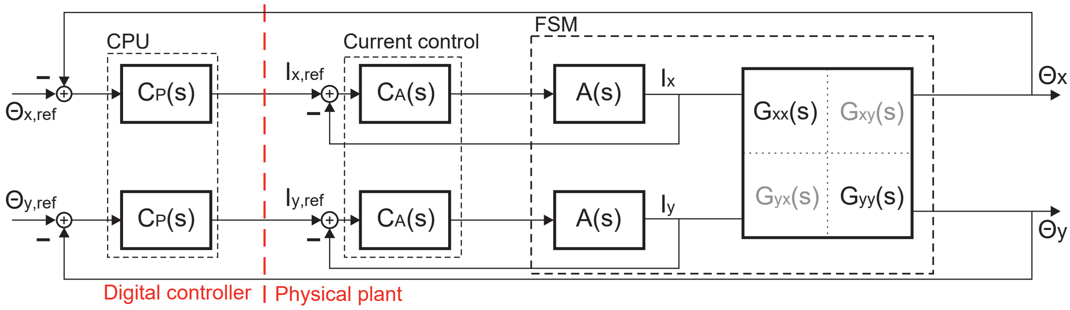


Figure 6: Control system.

current control cascade, the electrical subsystem can be neglected for the position control $C_P(s)$ for the targeted position control bandwidth ($\ll$4 kHz). The remaining part of the FSM can be modeled in the Laplace domain by

$$G_{xx}(s) = G_{yy}(s) = \frac{k_i}{js^2 + cs + k_\Theta + k_f} \tag{7}$$

with the motor-constant of the hybrid reluctance actuator k_i, the moment of inertia of the mover j, the damping coefficient c, the rotational stiffness of the flexure and the hybrid reluctance actuator k_f, k_Θ, respectively. Since the FSM is symmetrically designed, both axes are modeled by the same transfer function, which in turn allows the utilization of equal position control designs $C_P(s)$.

4 Electronics

4.1 Photosensitive device

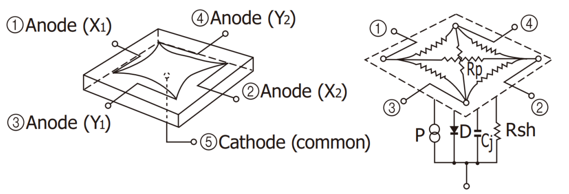
To sense the position of the laser beam in two dimensions, a photosensitive device (PSD) is used, manufactured by Hamamatsu type S5991. The principle of a two-dimensional PSD is based on the photoelectric effect with two anodes for each axis x/y and a common cathode, as depicted in Fig. 7a. The photo-electric effect causes a photo-current at each anode, which is depending on the position of the incident laser beam. The position of the laser spot on the PSD is given by the following equation:

$$x = \frac{L}{2}\frac{(i_{x2} + i_{y1}) - (i_{x1} + i_{y2})}{i_{x1} + i_{x2} + i_{y1} + i_{y2}}, \tag{8}$$

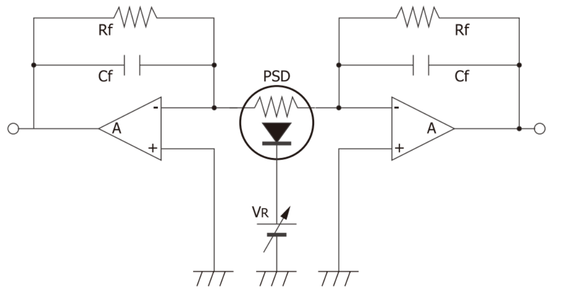
$$y = \frac{L}{2}\frac{(i_{x2} + i_{y2}) - (i_{x1} + i_{y1})}{i_{x1} + i_{x2} + i_{y1} + i_{y2}}, \tag{9}$$

with $L = 9$ mm being the length of the sensitive area and the photo-currents of each anode. The denominator normalizes the result to the sum current of all anodes.

Figure 7b shows an example of a circuit configuration used to retrieve the position signal of the laser beam in one axis. The readout circuit for the second axis is equivalent. The operational amplifiers are transimpedance amplifiers that convert each output currents of each axis $\mathbf{i_x}^T = [i_{x1} \quad i_{x2}]^T$ and $\mathbf{i_y}^T = [i_{y1} \quad i_{y2}]^T$ of the PSD into the output voltage signals



(a) Principle of the PSD with two anodes for each axis and a common cathode underneath.



(b) Sensor readout circuit for one axis of the PSD.

Figure 7: PSD principle and the readout circuit [3].

as,

$$\mathbf{u_x} = -R_f \cdot \mathbf{i_x}, \tag{10}$$
$$\mathbf{u_y} = -R_f \cdot \mathbf{i_y}, \tag{11}$$

where $\mathbf{u_x}^T = [u_{x1} \quad u_{x2}]^T$ and $\mathbf{u_y}^T = [u_{y1} \quad u_{y2}]^T$, are the output voltages of operational amplifiers, respectively. The capacitors C_f in the feedback of the operational amplifiers are added to attenuate high frequent noise of the photo-currents. The transimpedance amplifiers have an inverting behaviour, therefore each output voltage is inverted by an inverting amplifier. Additionally, a first order low-pass filter at the output of the PCB improves the noise suppression. These four analog signals are provided via ADCs to the digital controller (µC), where the calculation Eq. (9) of the position of the laser beam will be implemented, as visible in Fig. 9.

4.2 Current control

The electrical subsystem of the hybrid reluctance actuator is modelled by an inductance L_{HRA} and a resistor R_{HRA}. The dynamic behaviour of the electric circuit is described by

$$A(s) = \frac{1}{R_{HRA} + sL_{HRA}} \; . \tag{12}$$

To compensate the dynamic behavior of eq. (12), a proportional-integral (PI) controller

$$C_A(s) = k\frac{R_{HRA} + sL_{HRA}}{s} \tag{13}$$

is implemented. The factor k is tuned to reach a current control bandwidth of about 4 kHz. Fig. 8 shows the current control electronics with the reference current input on the left side and the current monitor output on the right side. Both voltage signals are proportional to the current and be transformed with $1\,\mathrm{A\,V^{-1}}$.

5 Experimental setup

An overview of the experimental setup used in the laboratory exercise is shown in Fig. 9. A laser diode is oriented toward the FSM, which directs the laser beam onto a PSD. The PSD

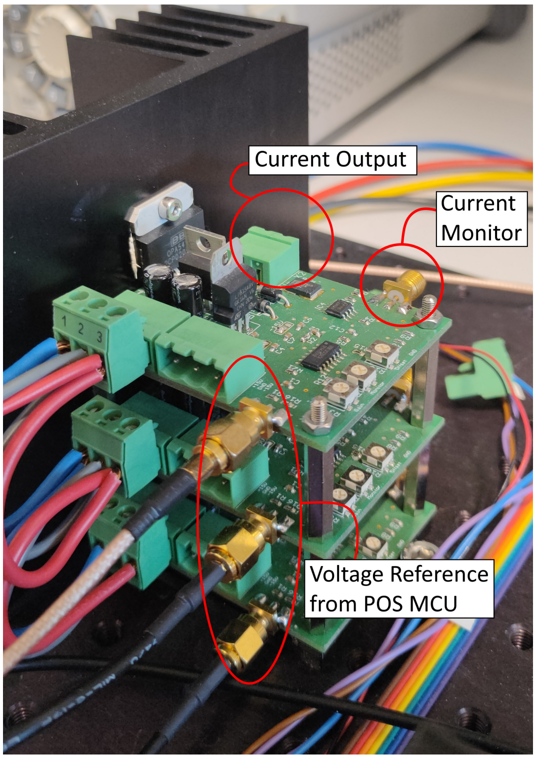


Figure 8: Current control electronics.

readout is used as the feedback position signal to control the FSM in two degrees of freedom. The PSD is mounted onto a 3D printed flexure, which can be actuated by a voice coil actuator (VCA) to later introduce disturbances along the vertical position.

5.1 Control hardware

Two STM32 microcontrollers (MCUs) for the processing of the laser position and for the position control are utilized. Both MCUs are equipped with two digital-to-analog converter (DAC) channels. These channels serve a dual purpose: they convey the laser's position from the first MCU to the second, and the second MCU uses them to define the reference for the current controllers for the FSM. The DACs operate within a voltage range of 0 V to 3.3 V.

The first MCU, designated as the *PSD MCU*, outputs the current laser position as an analog value for each axis via its DAC channels. The center position corresponds to 1.65 V, while the minimum and maximum values are limited by 0 V and 3.3 V, respectively. The analog position signals are required for the system identification and the position control. To provide the position signals the position control microcontroller, referred as *POS MCU*, connected to the corresponding ADC inputs.

The position control outputs are converted by the DACs, representing the reference for the current controllers for the x and y axis. These analog output signals are shifted by a custom op-amp circuit to convert the DAC voltage range of 0 V to 3.3 V to the symmetric range of $\pm$1.65 V, providing coils currents in a range of $\pm$1.65 A.

In the *SIMULINK* template files, some structures have already been defined. It is crucial to note that these processed signals require correct shifting and scaling for successful operation. *SIMULINK* utilizes a normalized value to describe the voltage range; specifically, 0 to 1 of the DAC output block is mapped to 0 V to 3.3 V. For example, a zero output current requires the POS MCU to set a *SIMULINK* signal value of 0.5, which corresponds to 1.65 V of the DAC and a final output voltage of 0 V with the op-amp post-processing.

To build and flash *SIMULINK* programs, the respective STM32 MCU must be connected via USB, and the correct MCU must be selected in the hardware settings of *SIMULINK* (either STM32H743ZI or STM32F767ZI).

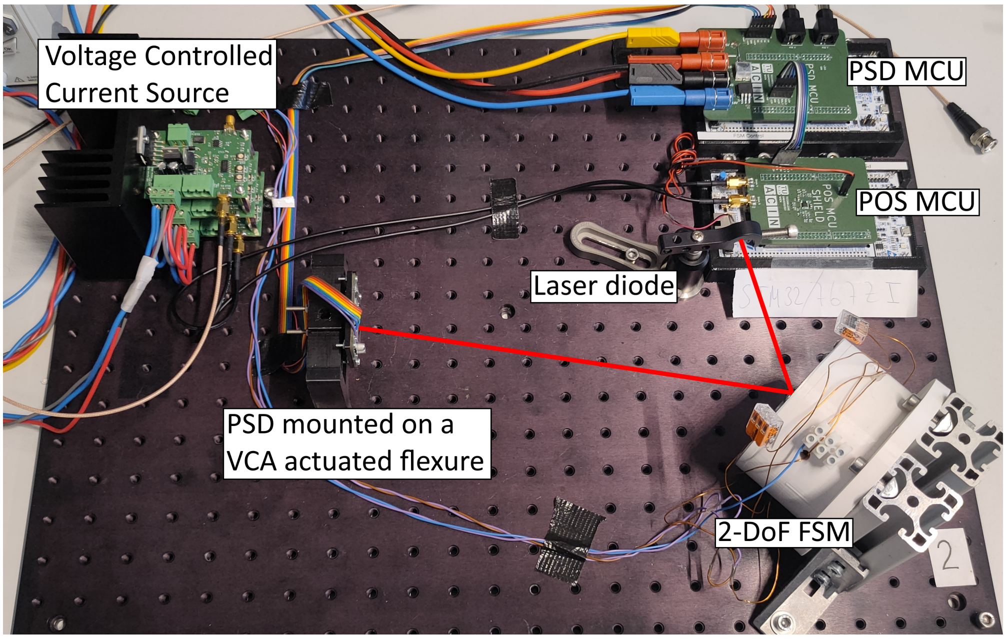


Figure 9: Overview of the experimental setup with the laser path highlighted in red.

5.2 System Identification

System identification will be performed using an Analog Discovery 3 (AD3) USB oscilloscope. A Bode plot of the system can be generated using the provided network analyzer tool within the Digilent Waveforms software, which is installed on the Lab PC. The AD3 features two waveform generator outputs, W1 and W2, and two analog inputs, CH1 and CH2.

To obtain a Bode plot trace, the "network analyzer" section in the software needs to be selected and configured appropriately. A sine wave is generated at the output W1, which must be fed back into CH1 while simultaneously connecting to the current amplifier board. The laser position output from the DAC of the PSD MCU needs to be fed into CH2. With these two signals, a Bode plot can then be traced.

5.3 Laser Diode and Current Amplifiers

The laser diode is supplied with 3.3 V from the MCU shield. The analog current controllers, depicted in Fig. 8, are supplied with ±18 V. They feature a reference signal input, which originates from either the AD3 or the STM32 boards depending on the task, and a monitor output to observe the current. The current control loop, as well as the gains and offsets on the controller PCB, are already tuned.

5.4 Voice Coil Disturbance

The voice coil actuator is used to introduce a disturbance along the vertical axis. The coil is driven with the aid of the current amplifier board. For the sensitivity function analysis, the coil needs to be driven with a signal from the AD3. The setup for this will be identical to that used for the system identification part, with the key difference being the actuation of the voice coil instead of one of the FSM axes.

6 Homework

a. Using the transfer function $G_{xx}(s)$ from Eq. (7), draw a bode-plot with the parameters $k_i = 1$, $j = 1$, $c = 0.1$, $k_\Theta = -1$ and $k_f = 2$ and add a time delay of of 10 ms.

b. Design a tamed PID controller $C_P(s)$ for the plant from [**a.**] with a -3dB bandwidth of higher than $\omega_{BW} = 10\,\mathrm{rad\,s^{-1}}$, and a phase/gain margin of at least 40° / 10 dB, respectively (See "rule of thumb" in [4]). Additionally, create a bode-plot with the plant $G_{xx}(s)$, the PID controller $C_P(s)$ and the open-loop transfer function $G_{xx}(s)C_P(s)$.

7 Exercise Instruction

7.1 Precautions

Failing to respect the following warnings may cause serious damage to the equipment and/or to the setup!

- **The output current of the power supplies must not not exceed ± 2 A!**
  Higher currents can cause the coils in the FSM or the voice coil actuator to overheat if the feedback control becomes unstable. The voltage of the controller output should not exceed ±18 V in this regard.

- **If the power supplies are in the current limit, turn them immediately off.**
  Otherwise the current amplifiers may be thermally destroyed.

- **Do not look into the laser!**
  The laser used is eye-safe, however it is not recommended to look directly into the laser diode.

7.2 Experiments

In the course of the following experiments, the system dynamics of the tip/tilt FSM, as well as, the cross-talk is examined. Based on the identified system dynamics, a high-dynamic position control for each DoF is tuned to stabilize the system. The closed-loop control performance is evaluated by an disturbed detector position.

Startup and derivation of PSD signal

1. As a first step, the derivation of the position signal has to be implemented. The photosensitive device (PSD) outputs four current signals, which allow to calculate the position of the laser spot on the PSD, as described by Eq. 9.

2. Implement these formulas in the *SIMULINK* template, using 4 ADC channels of the first microcontroller. Output the signals for the x- and y-axes on 2 DAC channels and verify them by using the AD3 oscilloscope.
3. Adjust the gain of the signals in *SIMULINK* to utilize the maximum voltage range to improve the signal-to-noise ratio.

System identification

4. Both FSM axes should be identified, as well as the cross-talk, using the AD3. Trace a bode plot for each axis and save the gathered data as a .csv file. The input is the reference signal input of the respective current controller for each FSM axis, the output is the DAC output of the position signal on the MCU shield. A MATLAB file is available which allows to convert the measured .csv file into a measured transfer function.
5. Start with one axis and trace a bode plot from 1 Hz to 5 kHz. Use small amplitudes at first to correctly measure the resonance of the respective axis.
6. Measure both axes and save the identified dynamics as a .csv file.

Control design

8. With the identified plant dynamics, design PID controllers for both axes. Choose a suitable bandwidth **above the resonance frequency** of the system on the mass line. Import the measured data into MATLAB and test (simulate) the controllers.
9. Plot simulated open-loop and closed-loop transfer functions with the measured plant and the designed controllers for both axes.
10. After verification, implement the controllers in *SIMULINK* on the STM32. Hint: You need to discretize the controllers using the sample time (40 µs) of the STM32. Set the reference in *SIMULINK* to 0 and observe the working controller by manually pushing the flexure with the mounted PSD.

Measurement of FSM performance

12. Turn off the controller. Trace a bode plot from 1 Hz to 1 kHz from the reference input of the voice coil actuator to the PSD signal output and save the acquired transfer function. Repeat the experiment with the active controllers. With the two measured transfer functions, calculate the sensitivity function.
13. Apply a sine signal with fixed frequency somewhere between 1 Hz and your chosen crossover frequency to the voice coil actuator. Measure the PSD signal output and discuss the resulting position signal error with working controllers.

7.3 Report

Write a report, including all the experimental results. Also, add the following points:

- Plot the measured bode plot of the system dynamics of both FSM axes and discuss the results. Are internal modes visible and how could they affect the operation of the system?
- Discuss the measured sensitivity functions. Up to which frequency is the implemented controller able to suppress disturbances of the voice coil actuator?

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